

Auteur: Rob Willemsen
Studierichting: Informatica
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$$\begin{aligned} & \int \frac{dx}{4x^2 + 12x + 18} \\ &= \int \frac{dx}{(2x + 3)^2 + 9} \\ &= \frac{1}{9} \int \frac{1}{\frac{(2x + 3)^2}{9} + 1} dx \\ &= \frac{1}{9} \int \frac{dx}{\left(\frac{2x + 3}{3}\right)^2 + 1} \\ &t = \frac{2x + 3}{3} \Rightarrow dt = \frac{2}{3} dx \Rightarrow \frac{3}{2} dt = dx \\ &= \frac{1}{9} \cdot \frac{3}{2} \int \frac{dt}{t^2 + 1} \\ &= \frac{1}{6} \operatorname{Bgtan}(t) + C \\ &= \frac{1}{6} \operatorname{Bgtan}\left(\frac{2x + 3}{3}\right) + C \end{aligned}$$

References